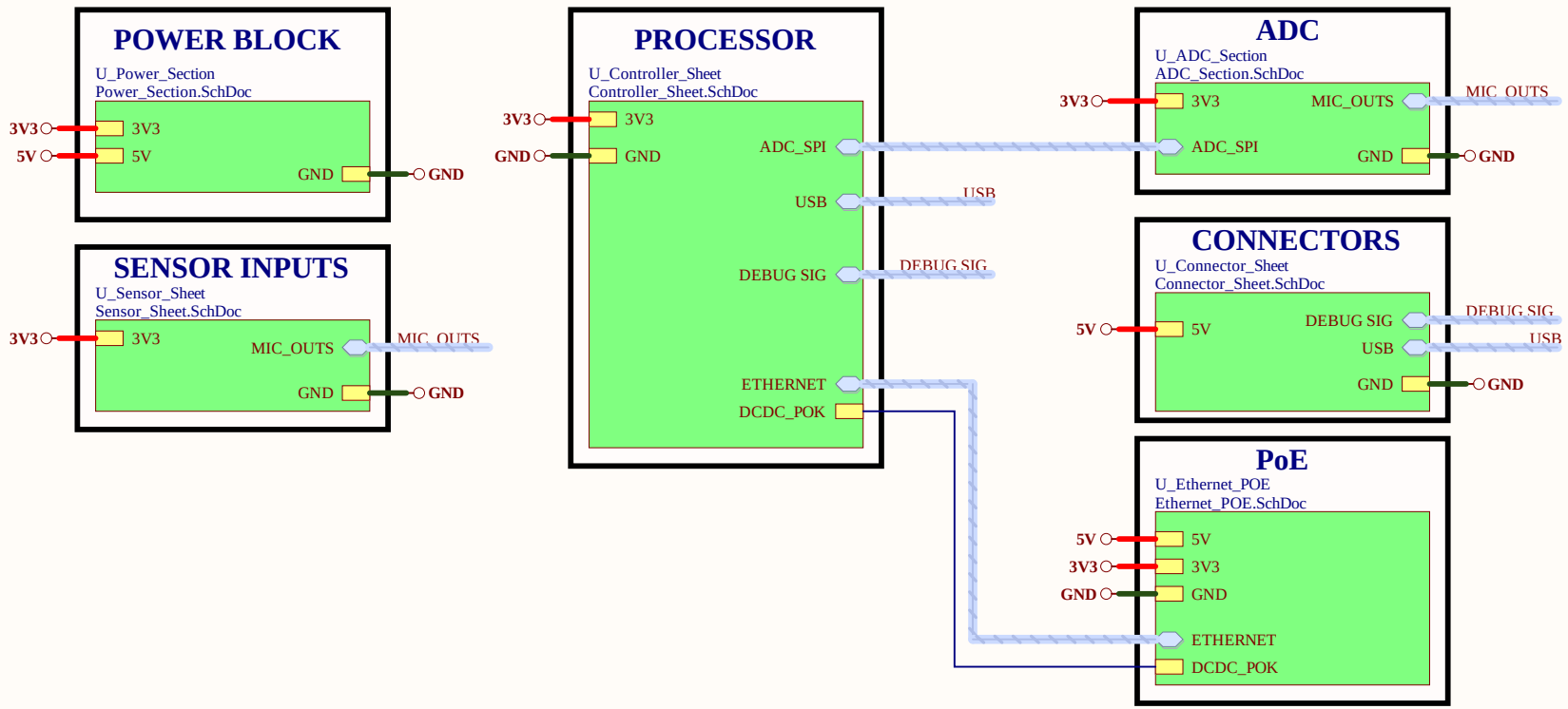
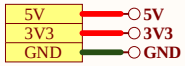
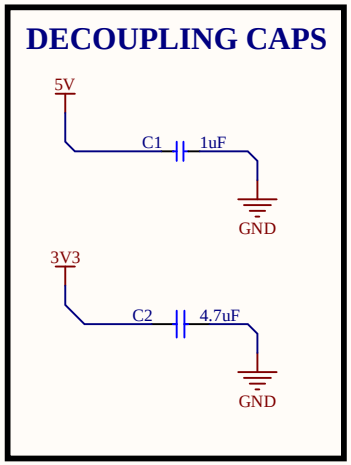
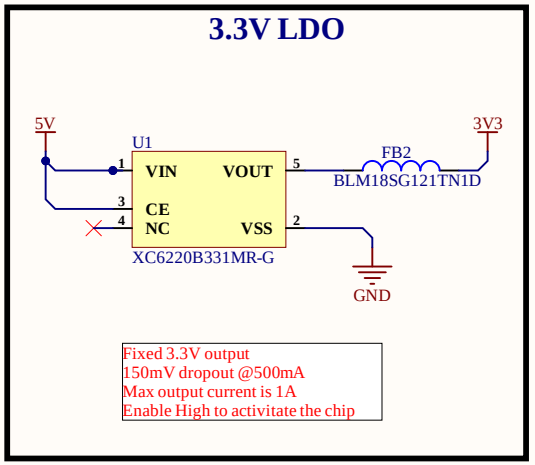


BLOCK_DIAGRAM

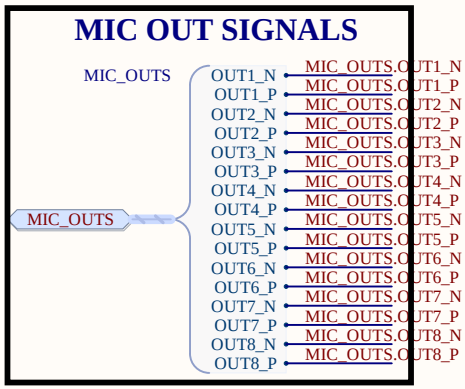


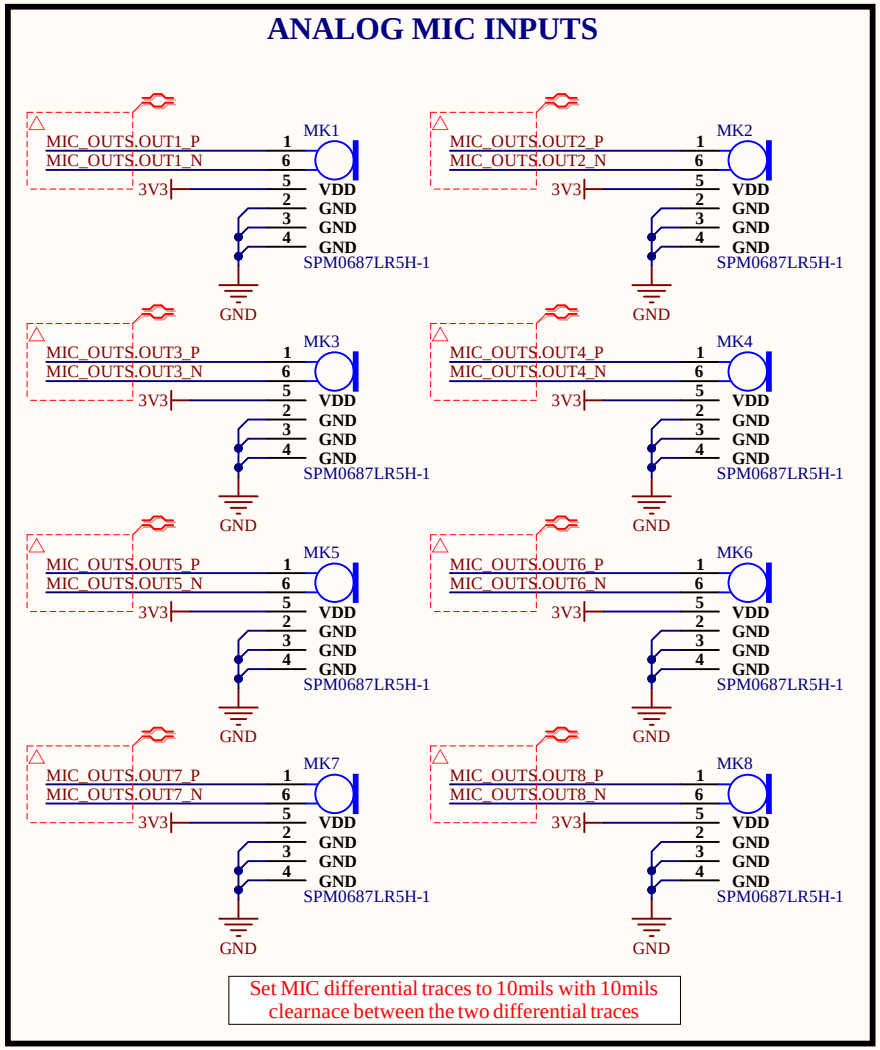


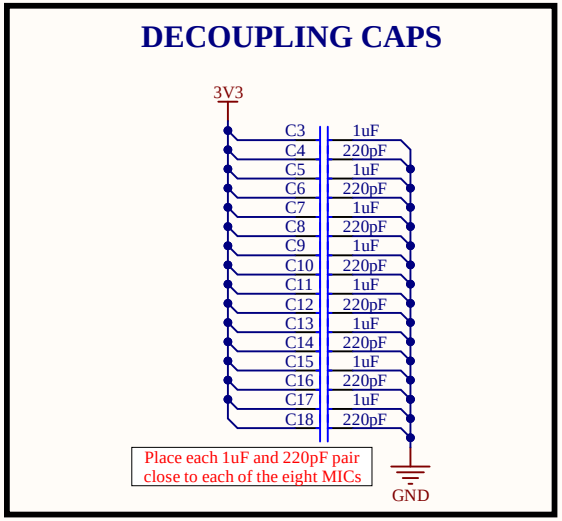


POWER_SECTION

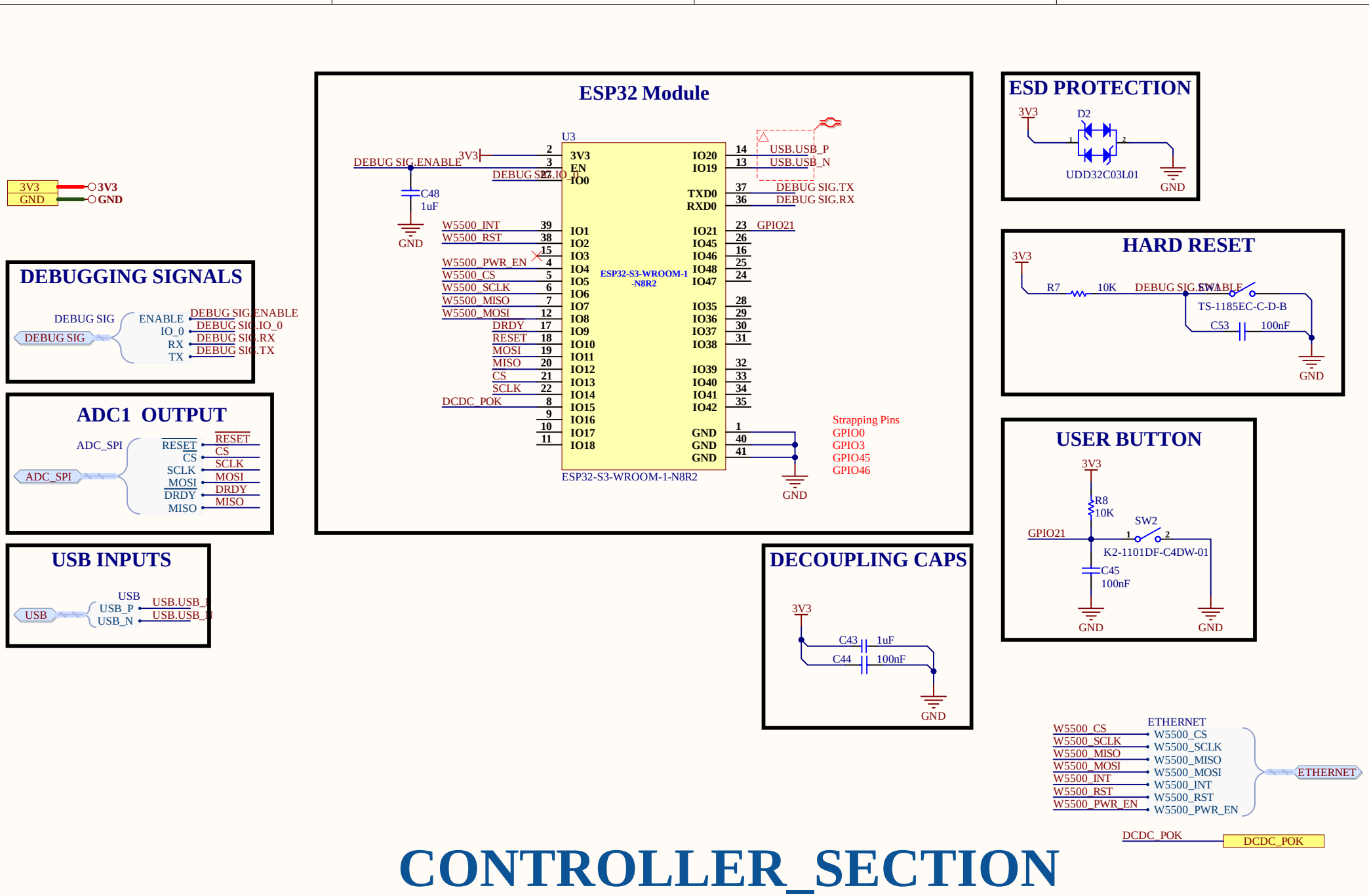


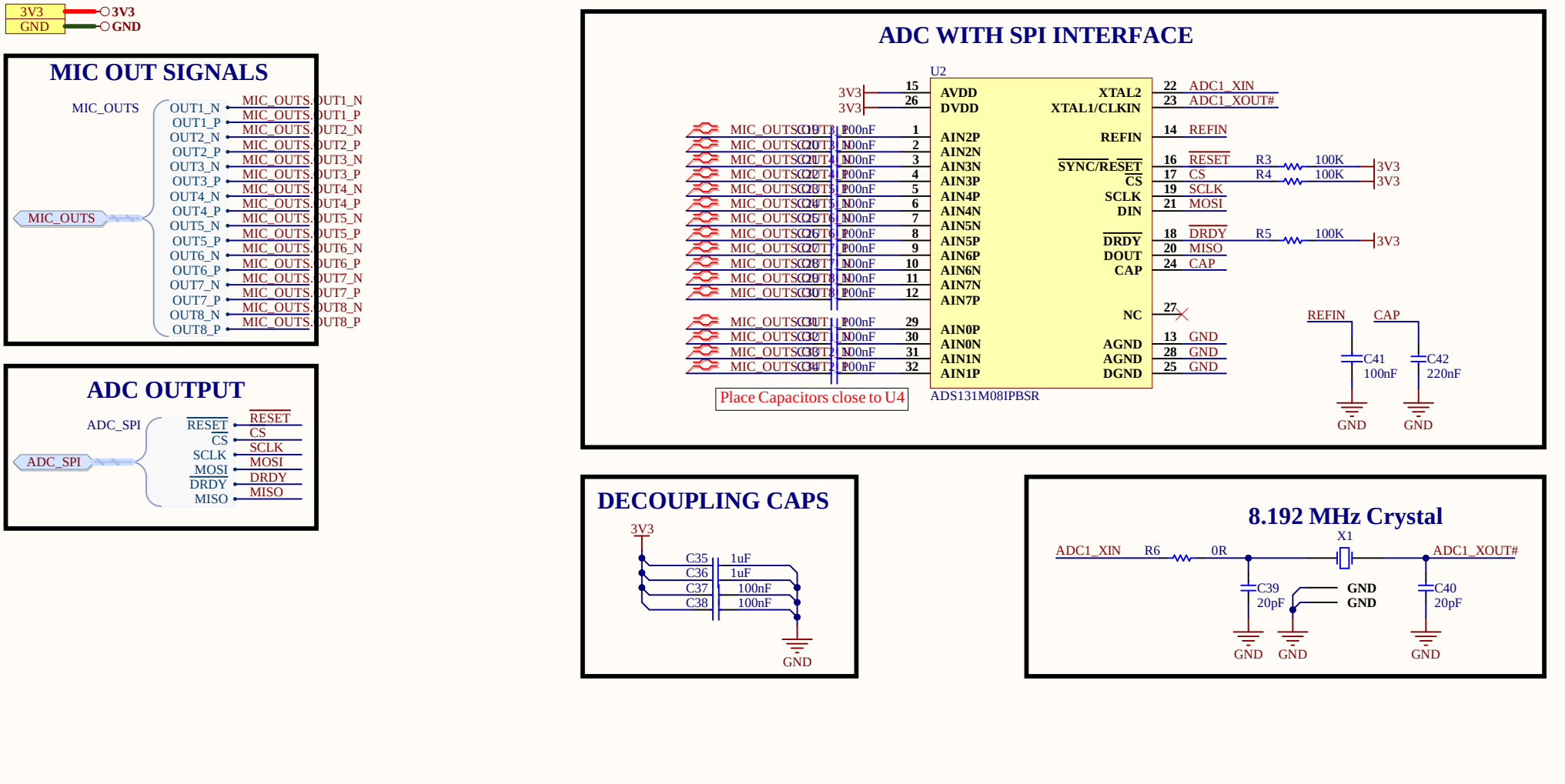






SENSOR_SECTION

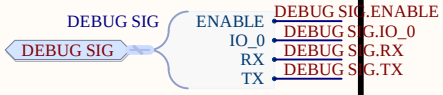

ESP32 Module
U3
ESP32-S3-WROOM-1-N8R2
DEBUG SIG.ENABLE
C48
1uF
W5500_INT
W5500_RST
W5500_PWR_EN
W5500_CS
W5500_SCLK
W5500_MISO
W5500_MOSI
DCDC_POK
DRDY
RESET
MOSI
MISO
CS
SCLK
USB.USB_P
USB.USB_N
DEBUG SIG.TX
DEBUG SIG.RX
GPIO21
Strapping Pins
GPIO0
GPIO3
GPIO45
GPIO46
ESD PROTECTION
D2
UDD32C03L01
HARD RESET
R7
10K
TS-1185EC-C-D-B
C53
100nF
USER BUTTON
R8
10K
SW2
K2-1101DF-C4DW-01
C45
100nF
DECOUPLING CAPS
C43
1uF
C44
100nF
DEBUGGING SIGNALS
ADC1 OUTPUT
USB INPUTS
ETHERNET
DCDC_POK
CONTROLLER_SECTION


3V3
GND
MIC OUT SIGNALS
MIC_OUTS
OUT1_N
OUT1_P
OUT2_N
OUT2_P
OUT3_N
OUT3_P
OUT4_N
OUT4_P
OUT5_N
OUT5_P
OUT6_N
OUT6_P
OUT7_N
OUT7_P
OUT8_N
OUT8_P
ADC OUTPUT
ADC_SPI
RESET
CS
SCLK
MOSI
DRDY
MISO
ADC WITH SPI INTERFACE
U2
AVDD
DVDD
AIN2P
AIN2N
AIN3N
AIN3P
AIN4P
AIN4N
AIN5N
AIN5P
AIN6P
AIN6N
AIN7N
AIN7P
AIN0P
AIN0N
AIN1N
AIN1P
XTAL2
XTAL1/CLKIN
REFIN
SYNC/RESET
SCLK
DIN
DRDY
DOUT
CAP
NC
AGND
DGND
ADC1_XIN
ADC1_XOUT#
R3
R4
R5
100K
C41
100nF
C42
220nF
Place Capacitors close to U4
ADS131M08IPBSR
DECOUPLING CAPS
C35
1uF
C36
1uF
C37
100nF
C38
100nF
8.192 MHz Crystal
X1
R6
0R
C39
20pF
C40
20pF
GND

ADC_SECTION



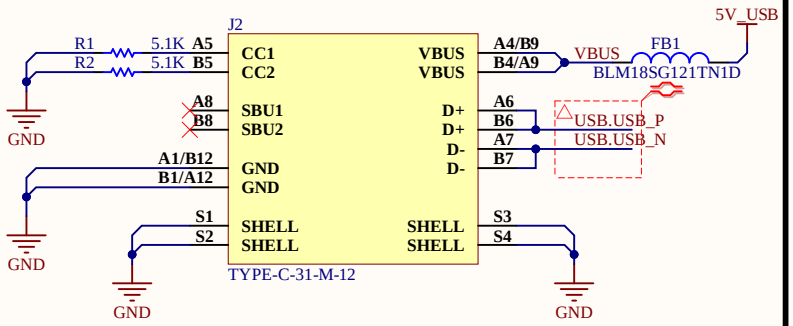
DEBUGGING SIGNALS



USB INPUTS



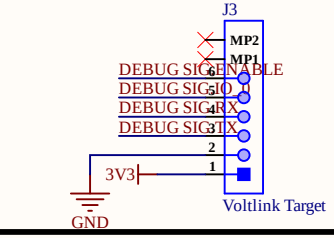
5V INPUT



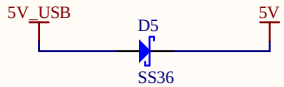
ESD PROTECTION



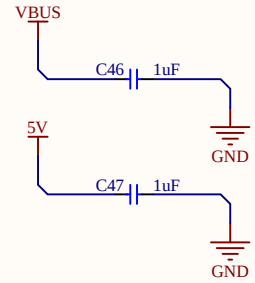
DEBUG CONNECTOR



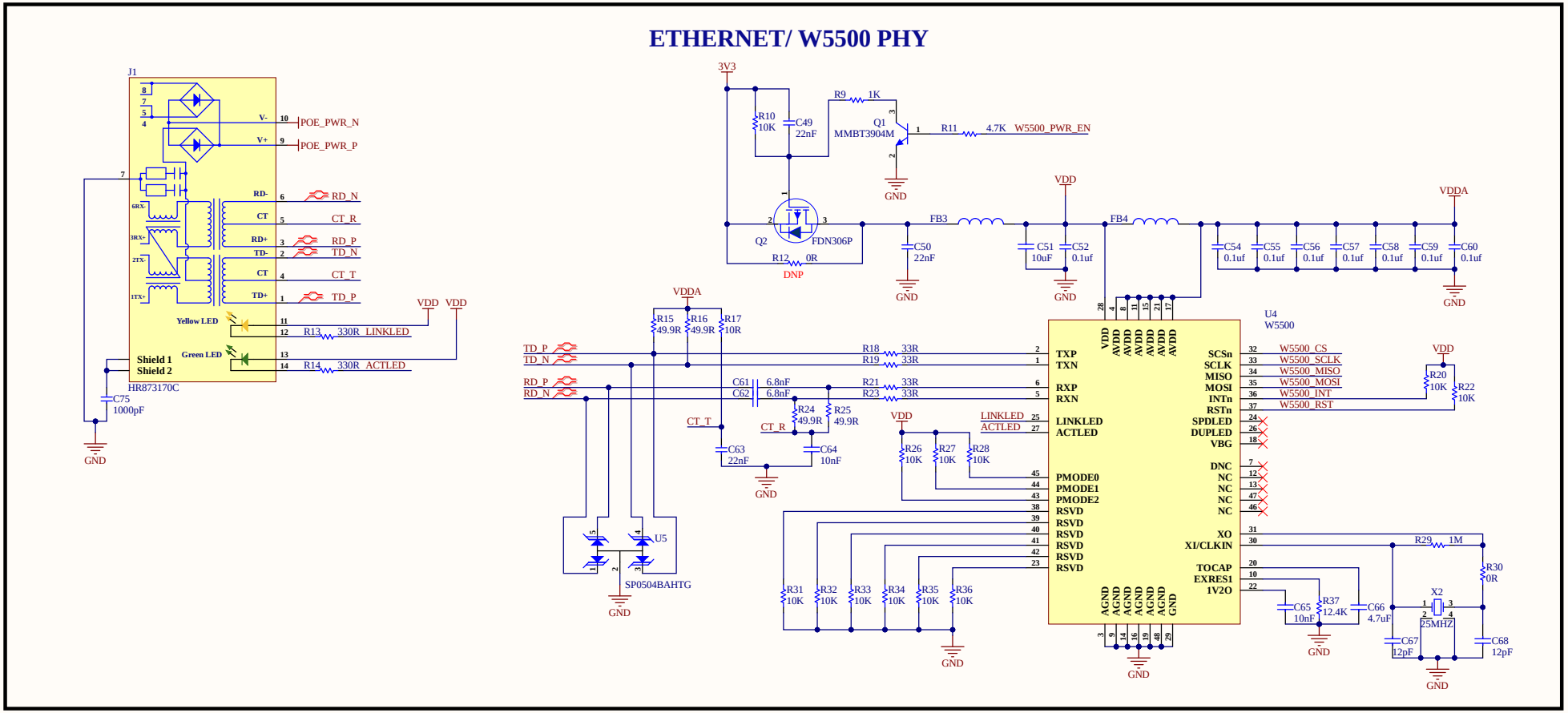
REVERSE POLARITY PROTECTION



DECOUPLING CAPS



CONNECTOR_SECTION


ETHERNET/ W5500 PHY
J1
HR873170C
POE_PWR_N
POE_PWR_P
RD_N
CT_R
RD_P
TD_N
CT_T
TD_P
Yellow LED
Green LED
R13 330R LINKLED
R14 330R ACTLED
C75 1000pF
3V3
R9 1K
R10 10K
C49 22nF
Q1 MMBT3904M
R11 4.7K W5500_PWR_EN
Q2 FDN306P
R12 0R DNP
FB3
FB4
VDD
VDDA
C50 22nF
C51 10uF
C52 0.1uf
C54 0.1uf
C55 0.1uf
C56 0.1uf
C57 0.1uf
C58 0.1uf
C59 0.1uf
C60 0.1uf
U4
W5500
R15 49.9R
R16 49.9R
R17 10R
R18 33R
R19 33R
R21 33R
R23 33R
C61 6.8nF
C62 6.8nF
R24 49.9R
R25 49.9R
C63 22nF
C64 10nF
R26 10K
R27 10K
R28 10K
LINKLED
ACTLED
PMODE0
PMODE1
PMODE2
RSVD
R31 10K
R32 10K
R33 10K
R34 10K
R35 10K
R36 10K
U5
SP0504BAHTG
SCSn
SCLK
MISO
MOSI
INTn
RSTn
SPDLED
DUPLED
VBG
DNC
NC
XO
XI/CLKIN
TOCAP
EXRES1
1V2O
W5500_CS
W5500_SCLK
W5500_MISO
W5500_MOSI
W5500_INT
W5500_RST
R20 10K
R22 10K
R29 1M
R30 0R
X2
25MHZ
C65 10nF
R37 12.4K
C66 4.7uF
C67 12pF
C68 12pF
POE
POE_PWR_P
POE_PWR_N
D3
SMAJ58A
C71 0.1uf
RDET
R39 24.9K
R42 180K
R43 RCLASS 4.42k
U6
TPS2375DR
DET
ILIM
CLASS
VSS
PG
RTN
NC
C70 15uF
R40 4.7K
R41 30mR
U7
TX4138
VIN
ILIM
EN
POK
BS
SW
FB
GND
R38 4.7R
C69 0.1uf
R44 DNP
C74 220pF
L1 33uH
D4 SS510B
C72 12pF
FB
R45 300K
Ra
Rb
R46 57.6k
F1
C73 470uF
POE_5V
R47 10K
DCDC_POK
VOUT = 0.805*(1+Ra/Rb)
REVERSE POLARITY PROTECTION
D6
SS36
5V
3V3
GND
ETHERNET
DCDC_POK
ETHERNET_POE